1 A hydroelectric power station could make a significant contribution to energy requirements.
Fig. 1.1 shows a dam at the hydroelectric power station which traps water behind a dam. When the height of the water behind the dam reaches 10.0 m, the water is released and passed through the turbines.

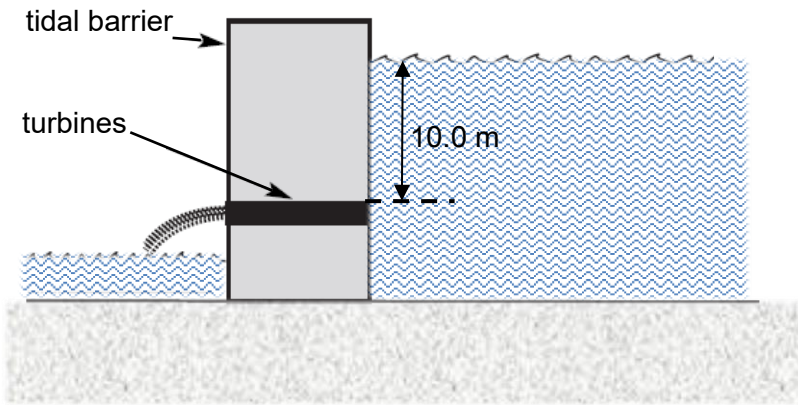

Fig. 1.1

(a) It takes 6.0 hours for a total mass of 1.32×10^{12} kg of water to flow through the turbines. The centre of mass of this amount of water is 5.0 m above the turbines.

Calculate the loss in the potential energy of the trapped water when it is released through the turbines completely. Assume that the density of the water is uniform.

loss in the potential energy = ……..J [2]

(b) The potential energy calculated in part **(a)** is lost and the efficiency of the power station is 40 %.

Calculate the average power output of the power station over this time period of 6.0 hours.

average power output = ……W [3]

[Turn over

(c) Suggest how the output power of the hydroelectric power station can be controlled as the level of trapped water decreases.

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2 In a binary star system, two stars, each of equal mass 3.5×10^{30} kg, rotate about their common